

AI Exploratory Test Report

Target Application: https://grand-occasion-pro.lovable.app

Report Date: March 24, 2026 at 11:16

Tool Version: AI Exploratory Test Engine v2.0

Pages Analyzed: 10

Total Test Cases Generated: 42

Valid Test Cases: 42

Rejected (Hallucinated): 0

Tests Executed: 42

Passed: 21

Failed (Bugs Found): 21

2. Execution Results Summary

21 PASSED

Tests behaved correctly

21 FAILED

Bugs found in application

Bugs Found - Failed Test Cases:

Test ID	Goal	Expected (Not Met - Bug)
VT-01	Trigger validation error on login page	validation failure for missing email
VT-03	Submit login form with empty email	validation failure for missing email
VT-06	Submit login form with valid email and empty on field	validation error for missing on field
VT-08	Test email input with very large number on root page	Invalid email address
VT-09	Test email input with very small decimal on login page	Invalid email address
VT-14	Trigger validation error on login page	Validation failure due to invalid email
VT-16	Check for error on empty email field	Error message for empty email field
VT-19	Test rapid input on login page	Error message for rapid input change
VT-21	Test multiple submissions on login page	Error message for duplicate submission
VT-23	Test clear and resubmit on login page	Validation failure due to previous submission
VT-24	Break mode switching on login page	No UI errors
VT-26	Test mode switching with incomplete form on login page	Validation failure

VT-27	Rapidly switch modes on register page	Error message
VT-28	Click mode button multiple times on login page	No UI errors
VT-29	Submit form after mode switch on register page	Error message
VT-30	Test mode switch with incomplete form on register page	Validation failure
VT-31	Rapidly apply and unapply mode on login page	Error message
VT-32	Stress test mode switching on login page	No UI errors
VT-33	Trigger error on empty email submission and verify erro	error message
VT-34	Test error recovery after invalid email submission	error clearance
VT-42	Test error recovery on root page after invalid email su	error clearance

3. Application Coverage Summary

Metric	Value
Pages Crawled	10
Pages with Interactive Elements	10
Input Fields Identified	12
Buttons Identified	12
Navigation Links	190
Dropdowns/Selects	0
Test Cases Generated	42
Validation Pass Rate	100.0%

4. Batch Generation Strategy

Tests were generated in 6 adversarial batches. Each batch focused on a distinct exploration category to guarantee structural diversity and maximize bug-finding potential.

Batch	Category	Focus
1	Input Torture and Field Poisoning	Edge-case values: long strings, special chars, min-1/max+1, empty required fields.
2	State Transitions and Form Abandonment	Partial fills, navigate-away-return, recalculate after change, state persistence.
3	Boundary Value Analysis	Exact min, max, zero, +1/-1 boundaries across every numeric input field.
4	Multi-Step Navigation Chaos	Fill on page A, jump to B, return to A -- does state survive?
5	Mode Switching and UI State Dances	Toggle modes mid-fill, switch tabs, check value retention after mode change.
6	Error Recovery and Validation Resilience	Submit empty, fix one field, resubmit; clear after success; recovery flow testing.

5. Test Coverage Classification

Category	Count
Authentication Tests	16
Negative Input Tests	8
Validation Enforcement Tests	6
General Exploratory Tests	5
Zero/Empty Input Tests	4
Boundary Tests	3

Test cases are automatically classified based on their goals and step patterns. Categories are not mutually exclusive; each test is assigned to its primary category.

6. Validated Test Cases

42 test cases passed anti-hallucination validation. All referenced UI elements exist in the application snapshot.

Each test is marked PASSED (green) or FAILED (red) based on execution results.

VT-01 | Trigger validation error on login page [FAILED - BUG FOUND]

Page: /loginTarget: Your emailType: Validation Enforcement Tests

1. Navigate to '/login'

2. Enter " into 'Your email'

3. Click 'Sign In'

Expected: validation failure for missing email

!! Application did not meet expected behavior - potential bug detected.

VT-02 | Test partial registration form submission [PASSED]

Page: /registerTarget: Your emailType: Authentication Tests

1. Navigate to '/register'

2. Enter " into 'Your email'

3. Click 'Create Account'

Expected: validation error for missing email

VT-03 | Submit login form with empty email [FAILED - BUG FOUND]

Page: /loginTarget: Your emailType: Validation Enforcement Tests

1. Navigate to '/login'

2. Enter " into 'Your email'

3. Enter 'on' into 'on'

4. Click 'Sign In'

Expected: validation failure for missing email

!! Application did not meet expected behavior - potential bug detected.

VT-04 | Trigger validation error on registration page [PASSED]

Page: /register

Target: on

Type: Validation Enforcement Tests

1. Navigate to '/register'

2. Enter " into 'on'

3. Enter 'test@example.com' into 'Your email'

4. Click 'Create Account'

Expected: validation error for missing on field

VT-05 | Test registration form submission with valid data [PASSED]

Page: /register

Target: Your email

Type: Authentication Tests

1. Navigate to '/register'

2. Enter 'test@example.com' into 'Your email'

3. Enter 'on' into 'on'

4. Click 'Create Account'

Expected: validation failure for duplicate email

VT-06 | Submit login form with valid email and empty on field [FAILED - BUG FOUND]

Page: /login

Target: Your email

Type: Validation Enforcement Tests

1. Navigate to '/login'

2. Enter 'test@example.com' into 'Your email'

3. Enter " into 'on'

4. Click 'Sign In'

Expected: validation error for missing on field

!! Application did not meet expected behavior - potential bug detected.

VT-07 | Submit registration form with valid email and empty on field [PASSED]

Page: /register

Target: Your email

Type: Validation Enforcement Tests

1. Navigate to '/register'

2. Enter 'test@example.com' into 'Your email'

3. Enter " into 'on'

4. Click 'Create Account'

Expected: validation error for missing on field

VT-08 | Test email input with very large number on root page [FAILED - BUG FOUND]

Page: /

Target: Your email

Type: Boundary Tests

1. Navigate to '/'

2. Enter '9999999999' into 'Your email'

3. Click 'Dismiss'

Expected: Invalid email address

!! Application did not meet expected behavior - potential bug detected.

VT-09 | Test email input with very small decimal on login page [FAILED - BUG FOUND]

Page: /login

Target: Your email

Type: Boundary Tests

1. Navigate to '/login'

2. Enter '0.0001' into 'Your email'

3. Click 'Sign In'

Expected: Invalid email address

!! Application did not meet expected behavior - potential bug detected.

VT-10 | Test on input with zero on register page [PASSED]

Page: /register

Target: on

Type: Boundary Tests

1. Navigate to '/register'
2. Enter '0' into 'on'
3. Click 'Create Account'

Expected: Invalid input

VT-11 | Test email input with exact minimum allowed value on about p [PASSED]

Page: /about

Target: Your email

Type: General Exploratory Tests

1. Navigate to '/about'
2. Enter 'a' into 'Your email'
3. Click 'Dismiss'

Expected: Invalid email address

VT-12 | Test email input with exact maximum allowed value on service [PASSED]

Page: /services

Target: Your email

Type: General Exploratory Tests

1. Navigate to '/services'
2. Enter 'a@a.a' into 'Your email'
3. Click 'Dismiss'

Expected: Invalid email address

VT-13 | Test email input with one below minimum on weddings page [PASSED]

Page: /services/weddings

Target: Your email

Type: General Exploratory Tests

1. Navigate to '/services/weddings'
2. Enter " into 'Your email'
3. Click 'Dismiss'

Expected: Email is required

VT-14 | Trigger validation error on login page [FAILED - BUG FOUND]

Page: /login

Target: Your email

Type: Negative Input Tests

1. Navigate to '/login'
2. Enter 'invalid_email' into 'Your email'
3. Click 'Sign In'

Expected: Validation failure due to invalid email

!! Application did not meet expected behavior - potential bug detected.

VT-15 | Test rapid submission on register page [PASSED]

Page: /register

Target: Your email

Type: Authentication Tests

1. Navigate to '/register'
2. Enter 'test@example.com' into 'Your email'
3. Click 'Create Account'
4. Click 'Create Account' again

Expected: Error message for duplicate submission

VT-16 | Check for error on empty email field [FAILED - BUG FOUND]

Page: /login

Target: N/A

Type: Authentication Tests

1. Navigate to '/login'
2. Click 'Sign In'

*Expected: Error message for empty email field***!! Application did not meet expected behavior - potential bug detected.****VT-17 | Test clear and resubmit on register page [PASSED]**

Page: /register

Target: Your email

Type: Authentication Tests

1. Navigate to '/register'
2. Enter 'test@example.com' into 'Your email'
3. Click 'Create Account'
4. Click 'Dismiss'
5. Enter 'new_test@example.com' into 'Your email'
6. Click 'Create Account'

*Expected: Validation failure due to previous submission***VT-18 | Trigger validation error on services page [PASSED]**

Page: /services

Target: Your email

Type: Negative Input Tests

1. Navigate to '/services'
2. Enter 'invalid_email' into 'Your email'
3. Click 'Dismiss'

*Expected: Validation failure due to invalid email***VT-19 | Test rapid input on login page [FAILED - BUG FOUND]**

Page: /login

Target: Your email

Type: Authentication Tests

1. Navigate to '/login'
2. Enter 'test@example.com' into 'Your email'
3. Enter 'new_test@example.com' into 'Your email'
4. Click 'Sign In'

*Expected: Error message for rapid input change***!! Application did not meet expected behavior - potential bug detected.****VT-20 | Check for error on missing required field [PASSED]**

Page: /register

Target: N/A

Type: Validation Enforcement Tests

1. Navigate to '/register'
2. Click 'Create Account'

*Expected: Error message for missing required field***VT-21 | Test multiple submissions on login page [FAILED - BUG FOUND]**

Page: /login

Target: Your email

Type: Authentication Tests

1. Navigate to '/login'
2. Enter 'test@example.com' into 'Your email'
3. Click 'Sign In'
4. Click 'Sign In' again

*Expected: Error message for duplicate submission***!! Application did not meet expected behavior - potential bug detected.**

VT-22 | Trigger validation error on about page [PASSED]

Page: /about

Target: Your email

Type: Negative Input Tests

1. Navigate to '/about'
2. Enter 'invalid_email' into 'Your email'
3. Click 'Dismiss'

Expected: Validation failure due to invalid email

VT-23 | Test clear and resubmit on login page [FAILED - BUG FOUND]

Page: /login

Target: Your email

Type: Authentication Tests

1. Navigate to '/login'
2. Enter 'test@example.com' into 'Your email'
3. Click 'Sign In'
4. Click 'Dismiss'
5. Enter 'new_test@example.com' into 'Your email'
6. Click 'Sign In'

Expected: Validation failure due to previous submission

!! Application did not meet expected behavior - potential bug detected.

VT-24 | Break mode switching on login page [FAILED - BUG FOUND]

Page: /login

Target: N/A

Type: Authentication Tests

1. Navigate to '/login'
2. Click 'on'
3. Click 'on'

Expected: No UI errors

!! Application did not meet expected behavior - potential bug detected.

VT-25 | Stress test dismiss button on register page [PASSED]

Page: /register

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/register'
2. Click 'Dismiss'
3. Click 'Dismiss'
4. Click 'Dismiss'

Expected: No UI errors

VT-26 | Test mode switching with incomplete form on login page [FAILED - BUG FOUND]

Page: /login

Target: Your email

Type: Authentication Tests

1. Navigate to '/login'
2. Click 'on'
3. Enter 'test@example.com' into 'Your email'

Expected: Validation failure

!! Application did not meet expected behavior - potential bug detected.

VT-27 | Rapidly switch modes on register page [FAILED - BUG FOUND]

Page: /register

Target: N/A

Type: Authentication Tests

1. Navigate to '/register'
2. Click 'on'
3. Click 'on'
4. Click 'Create Account'

*Expected: Error message***!! Application did not meet expected behavior - potential bug detected.****VT-28 | Click mode button multiple times on login page [FAILED - BUG FOUND]**

Page: /login

Target: N/A

Type: Authentication Tests

1. Navigate to '/login'
2. Click 'on'
3. Click 'on'
4. Click 'on'

*Expected: No UI errors***!! Application did not meet expected behavior - potential bug detected.****VT-29 | Submit form after mode switch on register page [FAILED - BUG FOUND]**

Page: /register

Target: Your email

Type: Authentication Tests

1. Navigate to '/register'
2. Click 'on'
3. Enter 'test@example.com' into 'Your email'
4. Click 'Create Account'

*Expected: Error message***!! Application did not meet expected behavior - potential bug detected.****VT-30 | Test mode switch with incomplete form on register page [FAILED - BUG FOUND]**

Page: /register

Target: Your email

Type: General Exploratory Tests

1. Navigate to '/register'
2. Click 'on'
3. Enter 'test' into 'Your email'

*Expected: Validation failure***!! Application did not meet expected behavior - potential bug detected.****VT-31 | Rapidly apply and unapply mode on login page [FAILED - BUG FOUND]**

Page: /login

Target: N/A

Type: Authentication Tests

1. Navigate to '/login'
2. Click 'on'
3. Click 'on'
4. Click 'Sign In'

*Expected: Error message***!! Application did not meet expected behavior - potential bug detected.**

VT-32 | Stress test mode switching on login page [FAILED - BUG FOUND]

Page: /login

Target: N/A

Type: Authentication Tests

1. Navigate to '/login'
2. Click 'on'
3. Click 'on'
4. Click 'on'
5. Click 'Sign In'

*Expected: No UI errors***!! Application did not meet expected behavior - potential bug detected.****VT-33 | Trigger error on empty email submission and verify error cle [FAILED - BUG FOUND]**

Page: /

Target: Your email

Type: Zero/Empty Input Tests

1. Navigate to '/'
2. Click 'Dismiss'
3. Enter " into 'Your email'
4. Click 'Dismiss'

*Expected: error message***!! Application did not meet expected behavior - potential bug detected.****VT-34 | Test error recovery after invalid email submission [FAILED - BUG FOUND]**

Page: /login

Target: Your email

Type: Negative Input Tests

1. Navigate to '/login'
2. Enter 'invalid' into 'Your email'
3. Click 'Sign In'
4. Enter 'valid@email.com' into 'Your email'
5. Click 'Sign In'

*Expected: error clearance***!! Application did not meet expected behavior - potential bug detected.****VT-35 | Verify error handling on registration with empty email [PASSED]**

Page: /register

Target: Your email

Type: Authentication Tests

1. Navigate to '/register'
2. Click 'Create Account'
3. Enter " into 'Your email'
4. Click 'Create Account'

*Expected: error message***VT-36 | Test error resilience on services page with invalid email [PASSED]**

Page: /services

Target: Your email

Type: Negative Input Tests

1. Navigate to '/services'
2. Enter 'invalid' into 'Your email'
3. Click 'Dismiss'
4. Enter 'valid@email.com' into 'Your email'
5. Click 'Dismiss'

Expected: error clearance

VT-37 | Trigger error on about page with empty email submission [PASSED]

Page: /about

Target: Your email

Type: Zero/Empty Input Tests

- 1. Navigate to '/about'
- 2. Enter " into 'Your email'
- 3. Click 'Dismiss'

Expected: error message

VT-38 | Verify error recovery on weddings page after invalid email s [PASSED]

Page: /services/weddings

Target: Your email

Type: Negative Input Tests

- 1. Navigate to '/services/weddings'
- 2. Enter 'invalid' into 'Your email'
- 3. Click 'Dismiss'
- 4. Enter 'valid@email.com' into 'Your email'
- 5. Click 'Dismiss'

Expected: error clearance

VT-39 | Test error handling on corporate page with empty email [PASSED]

Page: /services/corporate

Target: Your email

Type: Zero/Empty Input Tests

- 1. Navigate to '/services/corporate'
- 2. Click 'Dismiss'
- 3. Enter " into 'Your email'
- 4. Click 'Dismiss'

Expected: error message

VT-40 | Verify error resilience on birthdays page with invalid email [PASSED]

Page: /services/birthdays

Target: Your email

Type: Negative Input Tests

- 1. Navigate to '/services/birthdays'
- 2. Enter 'invalid' into 'Your email'
- 3. Click 'Dismiss'
- 4. Enter 'valid@email.com' into 'Your email'
- 5. Click 'Dismiss'

Expected: error clearance

VT-41 | Trigger error on concerts page with empty email submission [PASSED]

Page: /services/concerts

Target: Your email

Type: Zero/Empty Input Tests

- 1. Navigate to '/services/concerts'
- 2. Enter " into 'Your email'
- 3. Click 'Dismiss'

Expected: error message

VT-42 | Test error recovery on root page after invalid email submiss [FAILED - BUG FOUND]

Page: /

Target: Your email

Type: Negative Input Tests

- 1. Navigate to '/'
- 2. Enter 'invalid' into 'Your email'
- 3. Click 'Dismiss'
- 4. Enter 'valid@email.com' into 'Your email'
- 5. Click 'Dismiss'

Expected: error clearance

!! Application did not meet expected behavior - potential bug detected.

7. Structural Validation Confirmation

All 42 test cases were validated against the UI exploration snapshot. The validation engine verified that:

- ✓ All referenced UI element labels exist in the application snapshot.
- ✓ All referenced page paths exist in the crawled page inventory.
- ✓ No hallucinated or invented element labels were detected.
- ✓ Each test case has a valid structure: ID, goal, steps (min 2), and expected result.

8. Appendix

Snapshot Statistics

Metric	Value
Snapshot File	ai_exploration_snapshot.json
Snapshot Size	47208 bytes
Total Pages in Snapshot	10
Total Input Fields	12
Total Buttons	12
Total Navigation Links	190

Pipeline Configuration

Parameter	Value
AI Model	llama-3.3-70b-versatile (Grog)
Generation Batches	6
Tests per Batch	~10
Batch Delay	30 seconds
Validation Engine	3-Layer Anti-Hallucination
Step Generator	Deterministic (Template-Based)

Output Files Reference

File	Path
Raw Test Cases	data/generated_test_cases.json
Validated Test Cases	data/validated_test_cases.json

Rejected Test Cases	data/rejected_test_cases.json
Gauge Specification	specs/ai_exploration.spec
Step Implementations	step_impl/step_implementation.py
This Report	reports/test_report.pdf